

A System for Learning Atoms Based on Long Short-Term Memory Recurrent Neural Networks

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Abstract—In recent years, researchers in the fields of bioinformatics and cheminformatics have attempted to utilize machine learning methods for molecule modeling, bioactivity prediction, chemical property prediction, biology analysis, etc. In this paper, we present a system that merges the merits of various techniques such as long short-term memory (LSTM) recurrent neural networks, and is designed for learning atoms and solving the classic problems such as single task classification in the field of drug discovery. We have implemented our approach and conducted extensive experiments based on several widely used datasets such as SIDER and Tox21. The experimental results consistently demonstrate the feasibility and superiority of our proposed approach.

Index Terms—machine learning, drug discovery, neural networks, molecule data.

I. INTRODUCTION

Data-driven analysis plays a crucial part in many biological and chemical applications, including molecule modeling and chemical property prediction [1]. With the rapid development of machine learning [2], in recent years researchers in the fields of bioinformatics and cheminformatics have attempted to utilize machine learning methods for molecule modeling, chemical property prediction, biology analysis, and so on [3].

As we know, SMILES [4] (simplified molecular input line entry system) strings are usually used to represent and store molecule datasets, and they are in form of a single line text consisting of molecular notations. In realistic world, a molecule of arbitrary size and shape could be hard to be represented and used for machine learning tasks. Users usually need to transform them into other formats that are easy to be handled by machine learning algorithms. A widely adopted proposal is to use hand-crafted feature like ECFP [5], Coulomb Matrix [6], Graph-like structure [7], and so on. Such a process is usually called featurization. The transformed data (or featurization data) is usually as input and feed it into the interface of machine learning methods, such as classifier like random forest and multilayer perception, and so on [8].

Owing to the success of solving a wide range of machine learning problems by the Artificial Neural Networks(ANNs)

[9], recently, long short-term memory (LSTM) recurrent neural networks [10] have emerged as powerful generative models in various domains including natural language understanding [11] and images [12]. This lines of models regard the input data as sequential lists, and they are very suitable for solving time-dependent tasks like natural language understanding [13]. On the other hand, as shown in [14], the unsupervised machine, Atom2Vec [15], can learn the basic properties of atoms, and is used to discover the periodic table of the elements.

Inspired by the remarkable achievements mentioned above, in this paper we present a system that merges the merits of various techniques and is designed for solving the classic problems such as single-task classification and multi-task classification in the field of drug discovery. Generally, our system first transforms the molecule data in the SMILES format into a set of sample vectors via a component named format transformer (FT). The FT divides by spaces the molecule in SMILES format into atoms, which may consists of symbols and numbers. These atoms are then encoded by one-hot encoding, which allows us to transform atoms into a specific vectors with some certain dimensions. Then it uses a manner similar to word2vec [15] to extract the sample vectors by training the specific vectors previously obtained. These vectors are then as the input and fed into the embedding layer, which is used to extract high dimensional features. Meanwhile, in this layer a large matrix is constructed, which is convenient for model training at the upper layer (i.e., LSTM layer). The extracted features are then trained at the LSTM layer, and finally the trained samples are sent to a classifier (e.g., sigmoid) for single or multiple task classification.

In summary, the contributions of this paper are twofold: (i) we present an approach to learn atoms and solve the classic problems including single- and multi-task classification in the field of drug discovery; and (ii) we conduct extensive experiments based on several widely used molecule datasets and demonstrate the feasibility and superiority of our proposed approach. The rest of the paper is organized as follows. Section II reviews prior works most related to ours. Section III presents our approach. Section IV analyzes and discusses

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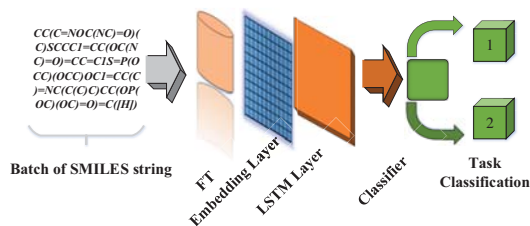


Fig. 1. The overall architecture of our approach

the performance results of our approach. Section V concludes this paper.

II. RELATED WORK

In the past decade, deep neural networks (DNN) has gained remarkable achievements in various areas of machine learning research. Deep neural networks can learn the potential regular pattern with access of training a large number of datasets to obtain better performance analysis and prediction. Different from other domains, DNN in drug discovery depends heavily on molecular featurization. The main molecular representation methods at present are ECFP [5], Coulomb Matrix [6], and Graph-like structure [7], and so on. Recently, deep neural models have opened up new avenues for modeling SMILES strings as a language model. The unsupervised machines (Atom2Vec) shown by [14] can learn the basic properties of atoms. Our work is inspired by theirs, yet it is different from theirs. In that paper the methods and experiments are used to discover the periodic table of the elements. In addition, they mainly focused on the principle explanation of atom representation, while related model designs are not covered. Another unsupervised approach named conditional diversity networks [16] also transform SMILES string into vector, while the detailed steps are not covered, and they paid more attention on the generation of molecule and drug, instead of the tasks mentioned in this paper.

On the other hand, there are some researches concentrated on drug predictions including adverse drug events (ADEs), adverse drug reactions (ADRs), adverse drug-drug interactions (DDIs), and so on. For example, Page *et al.* [17] identified the adverse drug events by relational learning. In addition, several methods proposed by [18] are for extracting ADRs from forum posts and tweets. As for adverse drug-drug interactions (DDIs), most of methods focused on binary prediction (with or without DDI). Jin *et al.* [19] formulated such a problem as a multitask dynamic regression problem. Dietterich [20] introduced the dynamic reposing technique that iteratively learns a neural network, and then focused on maximizing the predicted output values. Our work shares a common feature with this line of works, since both discussed the issues related to drug discovery. Nevertheless, our work is different from their in two points at least: (i) they mainly focused on developing techniques for other tasks instead of single- and multi-task classification; and (ii) the LSTM recurrent neural networks (RNNs) are not covered in their works.

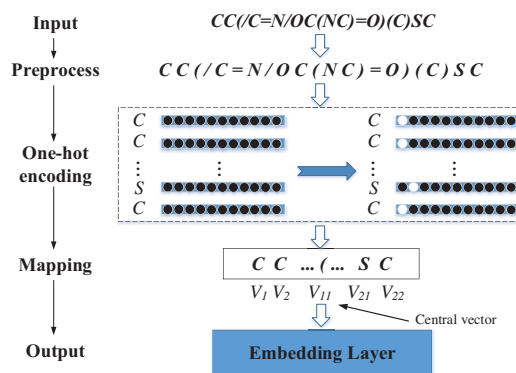


Fig. 2. The general process of FT

III. APPROACH

In this section, we first give an overview of our approach, and then present each part of our approach in detail.

A. Overview

Figure 1 shows the overall architecture of our approach. Briefly, the molecule data in the format of SMILES is first processed by a component named format transformer (FT), which transforms the molecule data into a set of sample vectors. These vectors are then as the input and fed into the embedding layer, which is used to extract high dimensional features. The extracted features are then trained at the LSTM layer using the long short-term memory (LSTM) recurrent neural networks. The output of the LSTM layer shall be processed by the classifier, which is used to generate the output label for task classification.

B. Representation

Choosing a proper molecular representation is at the heart of computer-based chemical analysis, and it is also very importance for drug discovery and prediction, since one may need to analyze and predict properties of drug with the same or similar molecule representation.

In the real world, most biological and chemical datasets are in the format of SMILES string. The SMILES string of a unique molecule is a single line text representation. For example, a molecule is encoded as a linearly arranged string $s_i = s_1, s_2, \dots, s_i (i = 1, 2, \dots, n)$. The encoding rules of SMILES follows the strict grammars, which consists of symbols indicating element types, bond values, and the start and terminal location for ring closures and branching components.

SMILES strings are powerful for representing and storing molecule data. To apply machine learning methods for learning advanced features, we need to transform them into the new format suitable for utilization. Take *Aldicarb* and its SMILES string CC(/C=N/OC(NC)=O)(C)SC as an example, one can convert it by RDKit [21] software into a graph-structured representation, which can be later used for learning features via graph convolutions. Instead of using chemical software such as RDKit to transform SMILES strings, we

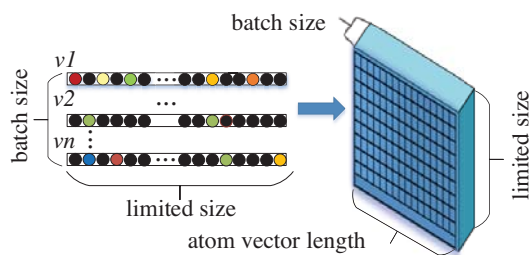


Fig. 3. The workflow of embedding implementation.

adopt another manner that directly transforms them into atom vectors. Briefly, molecule in SMILES format is first preprocessed as an independent atom symbol, and then they are expressed as a high dimensional vectors, which are sample vectors and also machine-readable characters or strings. Figure 2 illustrates the general process of this transformation. Next, we present the details of our transforming method, named Format Transfer (FT).

C. Format Transformer

As we known, in natural language processing (NLP), the sentences are processed using word vectors. We observe that SMILES is a linguistic grammar that employs an alphabet of characters to describe molecule, and each element or symbol has an associated definition in SMILES. Here we use the similar way in NLP to handle the SMILES strings. Specifically, for a series of molecules $s_i = s_1, s_2, \dots, s_i (i = 1, 2, \dots, n)$ in the format of SMILES, we divide them into a series of atoms by space. Each single atom x_i may consists of symbols and numbers. Then, for all preprocessed atoms, we encode them by one-hot encoding [22], which allows us to transform atoms into a specific vectors $v_i (i = 1, 2 \dots n)$ with some certain dimensions. Since these specific vectors may have less feature information. To obtain the sample vectors, we use a manner similar to word2vec [15] to extract the sample vectors, by training the specific vectors previously obtained. The sample vectors will be used as the input of the embedding layer.

Before the extraction process (i.e., training the specific vectors), we suggest a technique named simplified feature learning (SFL). This is based on the follow observation: molecules in SMILES format consist of numbers and characters, while some symbols and numbers may represent repetitive information, which may cause the complicated training process. For example, *Toluene* is denoted as Cc1ccccc1 in SMILES, a benzene ring is represented by number "1", while *c* and *C* denote aromatic and aliphatic carbon atoms, which essentially imply the existence of a benzene ring. Therefore, our SFL ignores some feature information that has been already expressed, and adds the occurrence frequency of each element (as the additional information) to the specific vector. These strategies ensure the simplicity and integrity of the featuring information.

D. Embedding Layer

When the sample vectors as the inputs are fed into the embedding layer, a huge matrix will be constructed, which

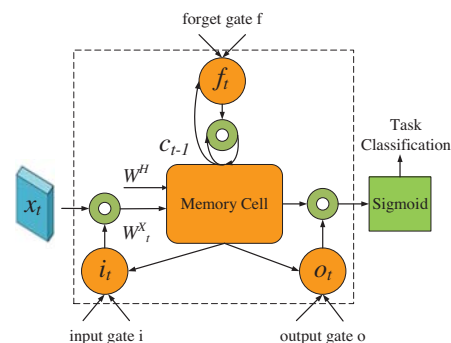


Fig. 4. An illustration of LSTM Memory Cell.

is convenience for model training in the later steps. The size of the matrix depends on the product of the batch size and the limited size (i.e., the dimension of an atom vector). Note that, every vector v_i , which is encoded using an N-bit status register. Each state has its own register bit, and at any time only one of them is valid. A workflow of embedding implementation is shown in Figure 3. The construction of the embedding matrix is based on the following principle: the element v_i located in the center of window k will be output object, and the other elements are the input ones (for example, see Figure 2, if atoms $v_i^* (i = 1, 2, \dots, 22)$ located in window k are as input, then the atom $v_i (i = 11)$ located in the central of window k will be output). In this way, it guarantees the output transmission channel unobstructed and ordered. In addition, each vector v_i encoded by one-hot encoding shall automatically find the corresponding vector in the pretrained matrix, until the mapping process completes.

E. LSTM Layer

Similar to the methods for dealing with semantics similarity in NLP, we use the long short-term memory (LSTM) recurrent neural network [10]. The LSTM is an alternative RNN, it uses the so-called memory cell (controlled by input, output and forget gates) to replace the conventional neuron in order to overcome the vanishing gradient problem of traditional RNNs. See Figure 4 for an illustration. In short, LSTM is a special class of RNN [23] that is capable of capturing long sentence relationships.

Owing to the existence of the gate of the adopted model, we can learn and recognize the information needed to be retained or forgotten. In our approach, each atom including special symbols (e.g., =, ≡), has a corresponding time step $x_t (t = 1, 2 \dots n)$. The intermediate state associated with each time step is referred to as a hidden state vector h_t . This hidden state vector is used to encapsulate and summarize all the information appeared in the previous time step. The hidden state is a function of the current atom vector and the hidden state vector of the previous step. The potential value of the hidden state vector would be

$$h_t = \sigma(W^H h_{t-1} + W^X x_t) \quad (1)$$

TABLE III
ROC-AUC SCORES OF EVERY TASK IN SIDER. THE RESULTS ARE BASED ON THE STRATIFIED SPLIT METHOD.

Task	RF		SVM		Ours	
	test	validate	test	validate	test	validate
Hepatobiliary disorders	0.5654	0.5696	0.5553	0.5733	0.5843	0.6916
Metabolism and nutrition disorders	0.5490	0.5337	0.5083	0.5091	0.5345	0.5382
Product issues	0.4906	0.5087	0.5032	0.5000	0.5048	0.5120
Eye disorders	0.5034	0.4984	0.5071	0.5179	0.5087	0.5260
Investigations	0.4877	0.5185	0.5014	0.5179	0.5045	0.4917
Musculoskeletal and connective tissue disorders	0.5180	0.5234	0.5116	0.5111	0.5620	0.5355
Gastrointestinal disorders	0.5551	0.4962	0.4926	0.5385	0.5564	0.5686
Social circumstances	0.4828	0.5185	0.4918	0.4958	0.5170	0.5204
Immune system disorders	0.5558	0.5271	0.5024	0.4950	0.5248	0.5465
Reproductive system and breast disorders	0.5436	0.5863	0.5601	0.6101	0.5956	0.5689
Neoplasms benign, malignant and unspecified (incl cysts and polyps)	0.5363	0.5744	0.5388	0.5676	0.5396	0.5073
General disorders and administration site conditions	0.4922	0.4962	0.4826	0.4955	0.4929	0.5061
Endocrine disorders	0.5245	0.5015	0.4920	0.4959	0.5323	0.5301
Surgical and medical procedures	0.5228	0.5609	0.4955	0.4858	0.4960	0.5000
Vascular disorders	0.4976	0.4981	0.5036	0.5129	0.5011	0.5132
Blood and lymphatic system disorders	0.5373	0.5248	0.5036	0.5833	0.5408	0.5869
Skin and subcutaneous tissue disorders	0.5417	0.5335	0.4959	0.4915	0.5642	0.5407
Congenital, familial and genetic disorders	0.4713	0.5015	0.4971	0.4950	0.5000	0.5059
Infections and infestations	0.5139	0.5005	0.4989	0.5135	0.5148	0.5224
Respiratory, thoracic and mediastinal disorders	0.4893	0.4952	0.5122	0.5238	0.4954	0.5335
Psychiatric disorders	0.5282	0.5126	0.5073	0.5070	0.5378	0.5168
Renal and urinary disorders	0.5632	0.5514	0.5137	0.5234	0.5767	0.6394
Pregnancy, puerperium and perinatal conditions	0.4769	0.4885	0.4962	0.5288	0.4961	0.4885
Ear and labyrinth disorders	0.5617	0.5781	0.5000	0.4938	0.5012	0.4851
Cardiac disorders	0.5530	0.5871	0.4899	0.5000	0.5734	0.5104
Nervous system disorders	0.4890	0.5346	0.5417	0.5147	0.5147	0.5522
Injury, poisoning and procedural complications	0.5262	0.5333	0.4947	0.5145	0.5315	0.5546

sets are used for evaluating models/approaches/systems. In our experiments we use two split methods mentioned in [24]. That is, random splitting and stratified random splitting. Usually, when there is some bias, random splitting is used for data splitting. In contrast, stratified random splitting ensures that each subset contains the full range of labels. To keep the same benchmark, in our experiments, random and stratified (random) split method with a 8/1/1 ratio are adopted for BBBP and BACE datasets, while only stratified (random) split method is adopted for Tox21 and SIDER datasets (since random split exhibits poor performance for all these methods).

B. Single Task Classification

For single-task classification (i.e., on BACE and BBBP datasets), we train several machine learning algorithms as baselines on an identical single-task dataset. These algorithms include LR (Logistic Regression) [27], RF (Random Forest) [28], SVM (Support Vector Machine) [29], DT (Decision Tree) [30] and KNN (K Nearest Neighbor) [31]. Also, for datasets used for these baselines, we use random and stratified splitting method with a 8/1/1 ratio. The comparison results are shown in Table I. It can be seen that our system generally outperforms these compared methods in both single-task datasets (i.e., BACE and BBBP). In particular, compared with the strongest baseline RF, in average, our system achieves an improvement of about 5% in these two datasets. These results demonstrate the competitiveness of our system. In addition, we observe that, for these two data splitting methods (i.e., random and

stratified), there is no significant impact on our system. This indicates the stability of the system.

C. Multi-Task Classification

In order to comprehensively analyze the predictive performance of our system for multiple task classification, experiments at this stage are conducted on each task in Tox21 and SIDER datasets. In this set of experiments, we only show the results of RF and SVM methods, since they present better performance than other baselines for multiple task classification. Specifically, there are 12 and 27 tasks for Tox21 and SIDER respectively. The compared results on Tox21 are shown in Table II. It can be seen that, on the whole our system shows competitive performance on the Tox21 dataset. Specifically, on 10 out of 12 tasks, our system achieves the best performance compared against the baselines.

In addition, the results on the SIDER dataset are shown in Table III. We can see that, on 20 out of 27 tasks, our system achieves the best performance. Overall, our system is still competitive, although it is inferior to some approaches on several tasks. But as we known, low ROC-AUC scores appeared commonly in most models found in the literature. Our system obtains much more leading scores for most tasks, and reaches the best results on more than 75% of the whole tasks, which essentially imply that our approach can effectively learn the structure and related information between atoms and the corresponding contexts.

V. CONCLUSION

In this paper, we presented an approach for learning atoms and solving the problem of single and multiple task classification in the field of drug discovery. Our approach transforms the molecule data in the SMILES format into a set of vectors via a component named the format transformer, and then feed them into the LSTM networks for training the sample vectors. We conducted extensive experiments based on several widely used molecule datasets. The results demonstrated the feasibility and superiority of our proposed approach. In the future, we would like to further optimize our approach and solve other related problems in the fields of drug discovery and bioinformatics.

VI. ACKNOWLEDGMENT

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